LUKE THOMPSON

JUNIOR SOFTWARE ENGINEER · AI-NATIVE PRODUCT ENGINEERING

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Portfolio: xylardark.github.io/Portfolio-Studio/u/admin/roserocket · GitHub: github.com/XylarDark

PROFESSIONAL SUMMARY

Junior software engineer who builds with AI coding agents and then checks their output before it merges. Since February 2026 I've directed agents on HomeWorld, my solo Unreal Engine 5.8 / C++ project, and built Python tooling that verifies results against real state. I also ship React/TypeScript with Postgres (Portfolio Studio) and TypeScript tests (Shift, 83 Vitest files). Earlier: C# REST integrations and game UI (OutFusion), JS/SQL web and C++ firmware (TruckSail), and client software and QA (Evenica).

CORE SKILLS

Languages	TypeScript/JavaScript, Python, SQL, C#, C++, HTML/CSS
Web & data	React, Vite, REST APIs, Supabase (Postgres), SQL Server, MySQL
AI agents	Directing AI coding agents (planning, review gates, verification of agent output), JSON tool bridges
Testing & CI	Vitest, pytest, headless-Chrome checks, Git/GitHub, GitHub Actions, pull-request review

SELECTED PROJECTS

HomeWorld — AI-agent tooling and verification Feb 2026 – present

Independent (solo) · Unreal Engine 5.8, C++, Python · private repository, available on request

- Built an editor automation bridge: agents and scripts send a JSON `{command, args}` request and get structured JSON back (world info, asset checks, a per-row pass/fail probe); new command modules plug in.
- Built checks that refuse a false "done": a headless test runner that hard-fails on five causes of fake passes (unreadable report, wrong engine, zero tests run, stale report, stale binaries), each with a regression test, and a build wrapper with one machine-wide queue so parallel agents can't collide.
- Run a multi-agent workflow (research, design, implementation, testing roles) with written handoff rules and merge gates; source changes wait for my review.
- A tool-capability matrix gives every claim an evidence level from E0 (only claimed in a doc) to E5 (survived a fresh editor). 42 C++ automation test files and 150+ Python test files.

Portfolio Studio — media-résumé site Sep 2026 – present

Independent · React, TypeScript, Vite, Supabase · xylardark.github.io/Portfolio-Studio

- React/TypeScript portfolio product with Studio editing, a Supabase (Postgres) backend and per-employer pages, built with AI coding agents through pull requests I review; headless-Chrome checks at desktop and phone sizes, and GitHub Actions deploys each merge.

Shift — playable shop & delivery game Aug – Sep 2026

Independent · Phaser 3, TypeScript, Vite · Live: xylardark.github.io/ShiftGame

- Shipped a TypeScript browser/PWA game for desktop and phones with a fixed 60 Hz simulation and cached A* routing; 83 Vitest test files in the public repo.

EXPERIENCE

Game Developer · OutFusion · Hamilton, ON Feb 2022 – Dec 2022

- Implemented game UI/UX from the design docs; built and tested C# REST API integrations for player data; integrated and tested game UI across mobile and desktop; partnered on bug verification and release quality.

Software Developer · TruckSail · Ancaster, ON Feb 2020 – Jan 2021

- Wrote C++ firmware for truck ECU devices (STM32, CAN bus) with defect finding for reliability; built JS/Bootstrap UIs and optimized SQL; maintained GitHub workflows and did cross-browser QA.

Software Developer & QA Tester · Evenica · Burlington, ON May 2018 – Sep 2019

- Built and tested client software (Andrew Peller, Young's Market Company, MM Realty International): T-SQL, C#, Linux, and Laravel (MySQL, PHP, JavaScript/jQuery, Vue.js).

Delivery Driver (Uber, retail delivery) · Hamilton, ON Jun 2022 – present

EDUCATION

Diploma, Computer Systems Technology — Software Development · Mohawk College · Hamilton, ON · 2019

Valid Ontario Class G licence · Open to hybrid in Toronto (commuting from Hamilton; Mon / Tue / Thu in office)